EECS 16A Designing Information Devices and Systems I
Discussion 2A

1. **Gaussian Elimination**

Gaussian Elimination is a systematic procedure that a computer could follow for solving a large system of linear equations simultaneously. The augmented matrix is in the form:

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right] \tag{1}$$

In Gaussian Elimination, we perform three operations on the system of equations. None of these three operations change the solution set of the system of equations. These operations are:

(a) Multiplying an equation by a *nonzero* scalar constant.

(b) Adding a scalar constant multiple of one equation to another.

(c) Swapping two equations.

Gaussian Elimination Algorithm for augmented matrix (1):

Step 1: For $i = 1, 2, \cdots, m$

(i) If necessary, swap row i with a row below it, so that the leading entry in row i is as far left as possible.

(ii) Rescale row i so that its leading entry is equal to 1.

(iii) For rows $j = i+1, i+2, \cdots, m$, add to row j a scalar multiple of row i, so that the leading entry of row i has all zeros below it.

Step 2: For $i = m, m-1, \cdots, 1$

(i) Add to each row $j = 1, 2, \cdots, i-1$ a scalar multiple of row i so that the leading entry of row i has all zeros above it.

For the following systems, use Gaussian Elimination to solve the problem. Does a solution exist? Is it unique? If there are an infinite number of solutions, give the solution in the parametric form.

(a) Let the three variables be x_1, x_2, x_3. Solve the following augmented matrix form using Gaussian Elimination.

$$\left[\begin{array}{ccc|c} 3 & -1 & 2 & 1 \\ 0 & 0 & 2 & 1 \end{array}\right]$$

(b) True or False: A system of equations with more equations than unknowns will always have either infinite solutions or no solutions.

(c)

$$\begin{bmatrix} 2 & 0 & 4 \\ 0 & 1 & 2 \\ 1 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -3 \\ 3 \end{bmatrix}$$

(d)

$$\begin{bmatrix} 1 & 4 & 2 \\ 1 & 2 & 8 \\ 1 & 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$$

(e)

$$\begin{bmatrix} 2 & 2 & 3 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \\ 1 \end{bmatrix}$$

